

Chapter 16

Applications Of Inversive Geometry

Alternative Reasoning

An integral part of an overall mathematical culture of a mathematically-inclined thinker is the ability to generate as many distinct solutions of a single well-defined academic problem as possible, be it a problem on the theory of numbers, the theory of graphs, the theory of groups, real or complex analysis, classical/analytical/differential geometry and so on.

Intuitively speaking, such a process of developing multiple solutions to a single problem is, give or take, a converse of the process of developing a single method or a single machinery that is capable of solving multiple, possibly very distinct, problems. Which is a corner of mathematics of its own.

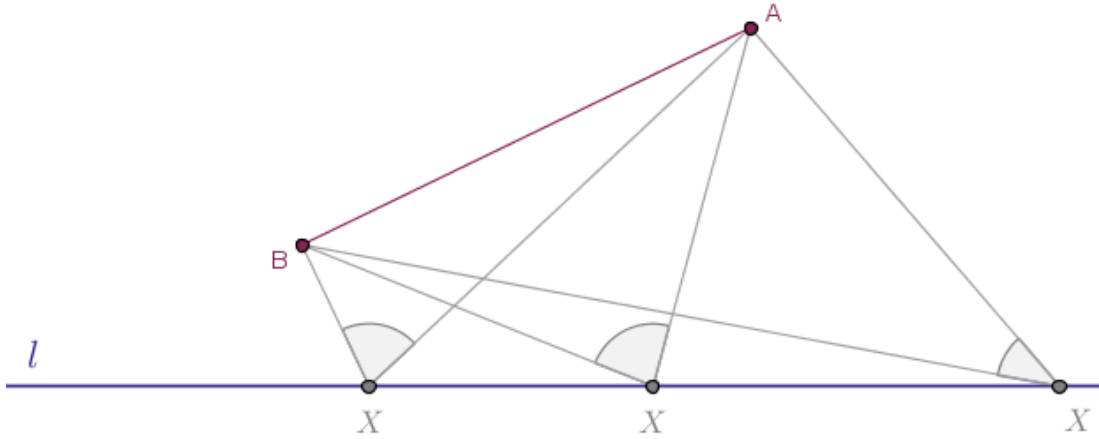
In the classrooms, however, the exercise of looking at a single problem through the lens of multiple, distinct, solutions is rarely, at best, and, more realistically and more sadly, is never undertaken. Such an activity is mostly the purview of the so called math circles, physics-mathematical schools, physics-mathematical preparatory programs (attended by the author of this book) and, hopefully, college seminars or recitations.

Psychologically it may be quite challenging to simply not drop such a solved problem on the floor cold turkey, step over it and move on. Who cares? Here is *a* simple solution that we have found already, what else do you want? But that is precisely the crux of the matter. We learn to do things that are *not* easy.

That is, developing the mathematical stamina of not giving up and raining on a technical problem bit by bit is something that will be useful in anyone's later mathematical career.

More immediately, juggling in one's mind various machineries of various, vaguely defined, distinct branches of mathematics to see which one may work, constitutes the very process of *studying* and *learning* mathematics.

Take for example, the geometric optimization problem of finding the location of a mystery point, X , on a given straight line, l , from which a fixed coplanar line segment, AB , is seen under a maximal angle (Figure 16.1):



There exist only two distinct *angular* mutual juxtapositions of the straight lines l and AB :

- the straight lines l and AB are either parallel (do not intersect)
- or they are not parallel (they do intersect)

Among those, the first mutual juxtaposition of the straight lines l and AB , when $l \parallel AB$, is not interesting because in that case the solution is trivial. The mystery point X on l is located in such a way that the triangle AXB is isosceles.

Which means that to actually locate the mystery point X on l all that needs to be done is the following: snap a perpendicular bisector, p_{ab} , of the line segment AB . The point where such a bisector intersects the given line marks the spot: $p_{ab} \cap l = \{X\}$. Game over.

As such, from now on we assume that the straight lines l and AB are not parallel: $l \nparallel AB$.

Under that assumption there exists only three mutual juxtapositions of the straight lines l and AB . Namely, the straight line l intersects:

- the line segment AB at a strictly interior point, O , such that $A < O < B$
- the line segment AB at a boundary point, A or B
- the straight line AB extended

In the first case above the intersection point, O , is the point X because from that point the line segment AB is seen under the angle of π radians or 180° , the largest angle possible under the circumstances.

In the second case above the angle between the straight line and the line segment AB is not even defined.

Hence, we now only consider the third case, the case depicted in Figure 16.1 above. A slightly less generic and a slightly more specific version of that problem is known as [the Angle Maximization Problem of Regiomontanus](#).

The wiki page of that problem, as of this writing circa March of 2026, shows three distinct solutions of the Regiomontanus problem based on:

- elementary geometry
- differential calculus
- high school algebra

The above solutions that are based on differential calculus and high school algebra are analytic in nature and do not paint any visual picture of the answer.

In contrast, the above solution that is based on elementary geometry is quite visual and it is constructible with Euclidean tools, a compass and a straight edge, alone. In particular, the sought-after point, X , is the point where a circle that passes through the given points A and B touches the given straight line l .

Thus, from the synthetic geometry perspective, the solution of the Angle Maximization Problem of Regiomontanus boils down to the solution of the so-called **LPP** or the Second Problem of Apollonius. That problems admits two distinct solution circles but in our case we only pick one such circle, the circle that generates the largest angle between the two.

That is, each one of the two solution circles of the **LPP** problem will generate two distinct points, say, X and Y , on the given straight line, l . Consequently, across the two angles AXB and AYB we are to pick the larger one. Which means that across two said solution circles we pick only one.

Separately, it can be straightforwardly shown* that between two of these circles for our angle maximization problem we pick **the smaller** one, the solution circle with the smaller radius.

Now that we have setup the context, we want to get very technical and zoom on a very specific and a highly tactical task of constructing the two candidate solution circles of the General Angle Maximization Problem depicted in Figure 16.1 using precisely the machinery of inversive geometry.

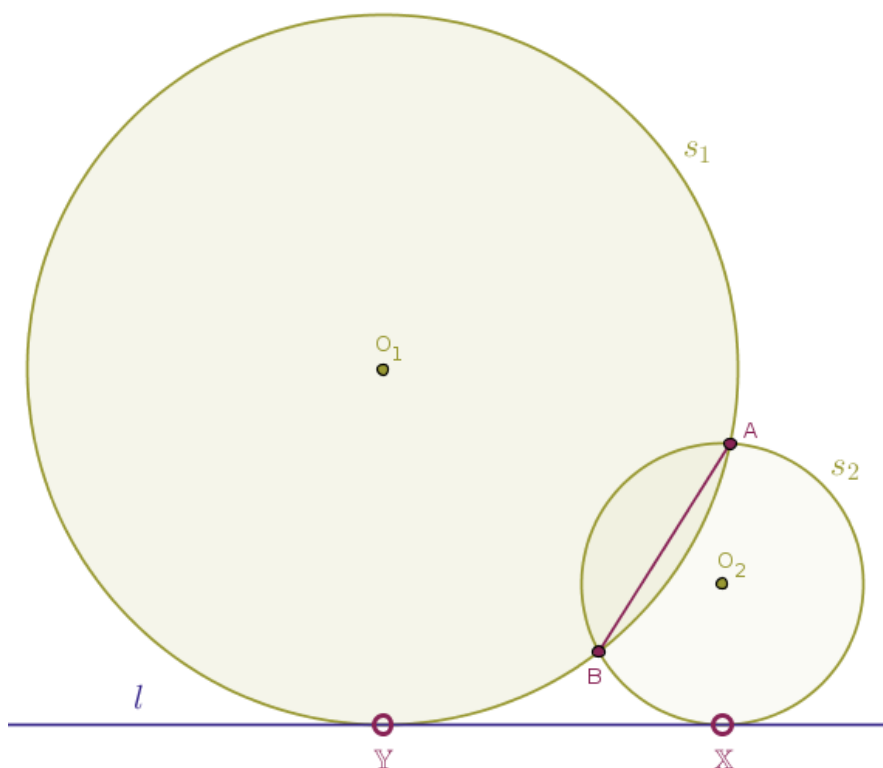
Yes, that specific problem can be solved in a number of different ways none of which employs inversive geometry but befitting the nature of this book, we, naturally, want to apply the newly learned material to the solution of various technical problems. As such.

Problem 22. Use the machinery of inversive geometry in order to construct two distinct circles that pass through the given points, A and B , and that touch the given straight line, l .

All the given objects are considered to be fixed, distinct and coplanar. Use the Euclidean tools only.

Solution. Solving this problem in reverse order, we temporarily assume that the mystery circles sought-after have been constructed somehow already. Exactly how these circles have been constructed is of no interest to us yet because at this stage of our solution we simply want to see if it is possible to extract any actionable mathematics from such an assumption.

To that end. We already know that the two mystery solution circles, $s_{1,2}$, sought-after must both pass through the points A and B , the points that delineate the given line segment, AB (Figure 16.2):

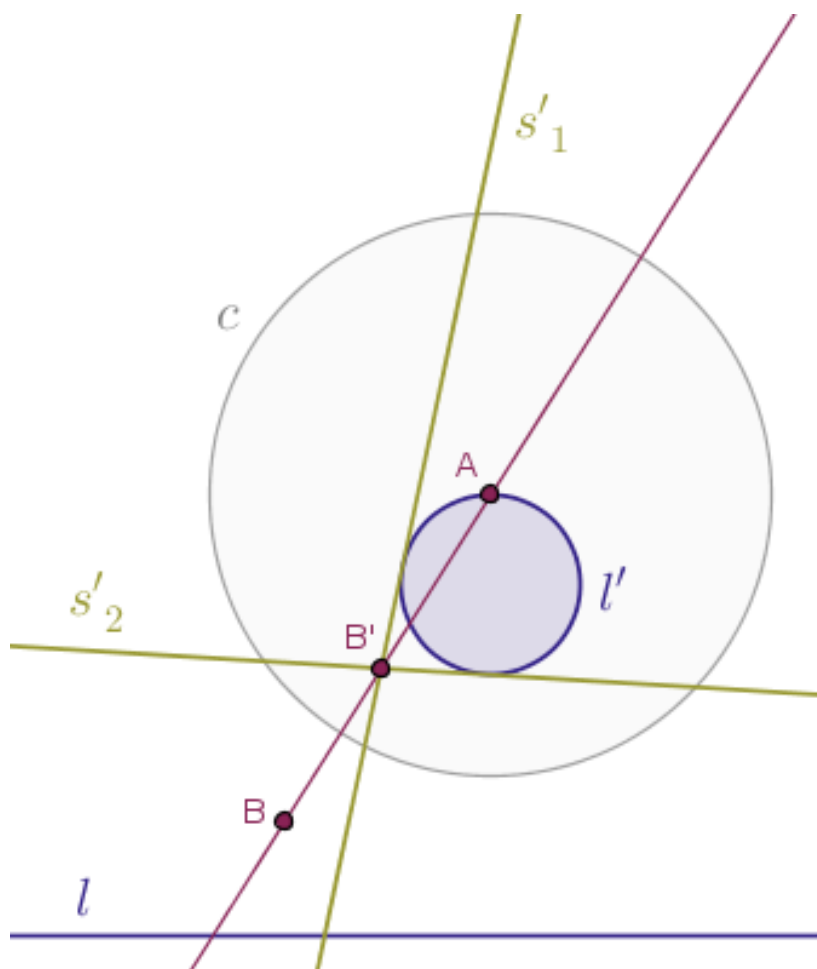


Wonderful. Now we recall the family of Theorems 15 through 19. According to these theorems any circle that passes through a center of inversion will be taken into a *straight line*.

But, when it comes to popular tangency problems, dealing with straight lines is usually much more tactically advantageous than dealing with curved lines or circles. Thus, we already feel that we are onto something here if we pick any point that delineates the given line segment, A or B , to be the center of inversion, with positive power (perhaps).

For demonstration purposes, let the point A be the center of inversion with positive power. Let us see where it will take us. Under the inversion of the parent plane in a circle $c(A, r)$ that has the point A as its center and any positive real number r ($r > 0$) as its radius, we understand, the two intersecting mystery circles $s_{1,2}$ will be taken into two intersecting straight lines, $s'_{1,2}$.

By the Theorems 11 through 14, the given straight line, l , that does *not* pass through the point A (by earlier assumption) will be taken into a circle, l' , that *does* pass through the center of inversion, the point A (Figure 16.3):



The straight line AB that, evidently, passes through the center of inversion, the point A , will be taken into itself by the Theorem 9 and the point B will be taken into a point B' .

Now. Where exactly did we win here? Our tactical win is the straightening of the curved objects $s_{1,2}$ into the straight lines $s'_{1,2}$. Why is that?

Because if the problem of constructing two curved objects, $s_{1,2}$, that must touch one straight line, l , is rated as *difficult* then the problem of constructing two straightened out objects, $s'_{1,2}$, that must touch one circle is rated as a previously solved problem, since we know how to construct two straight lines, $s'_{1,2}$, that are tangent to a given circle, l' , and that pass through a given point, B' .

Here it should be evident that if the mystery circles $s_{1,2}$ intersect at the point B then the images of these circles, the straight lines $s'_{1,2}$, will intersect at the image of the point B or the point B' and, fundamentally, we are done. We have solved the problem completely and we can frame the following alternative solution of the **LPP** problem of Apollonius:

Step 1. Choose the given point A as the center of inversion of the parent plane with respect to a circle $c(A, r)$ of radius r with positive power (Figure 16.4):



Step 2. Reflect the input straight line l in the circle $c(A, r)$ with positive power into a circle, l' .

Step 3. Reflect the given point B in the circle $c(A, r)$ into a point B' (with positive power).

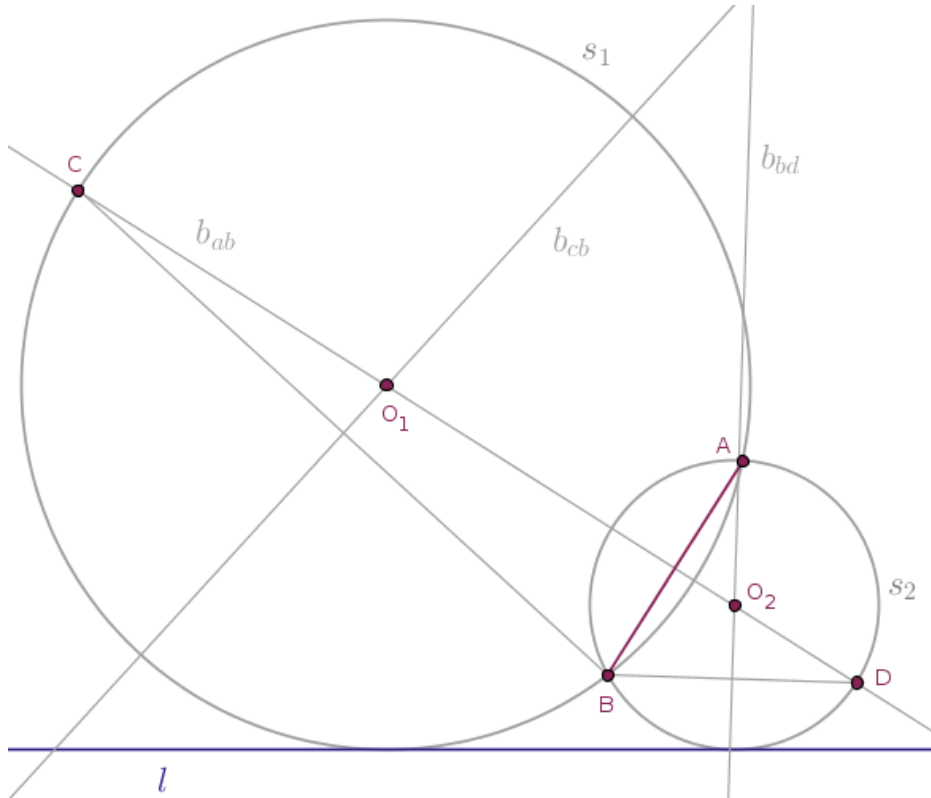
Step 4. Construct two straight lines, $s'_{1,2}$, that pass through the point B' and are *tangent* to the circle l' as is shown in Figure 16.3 above.

Step 5. Reflect the newly obtained straight lines $s'_{1,2}$ in the circle $c(A, r)$ into the target circles $s_{1,2}$. The two circles $s_{1,2}$ thus obtained touch the input straight line l and pass through the given points A and B .

What remains to be recovered now are the points X and Y over which the newly constructed circles $s_{1,2}$ touch the given straight line l . That task can be carried out in a number of different ways.

For example. If the points X' and Y' over which the straight lines $s'_{1,2}$ touch the circle l' are known, due to the very process of constructing the tangents $s'_{1,2}$, then it is always possible to simply reflect those points in the circle $c(A, r)$ into the sought-after points X and Y .

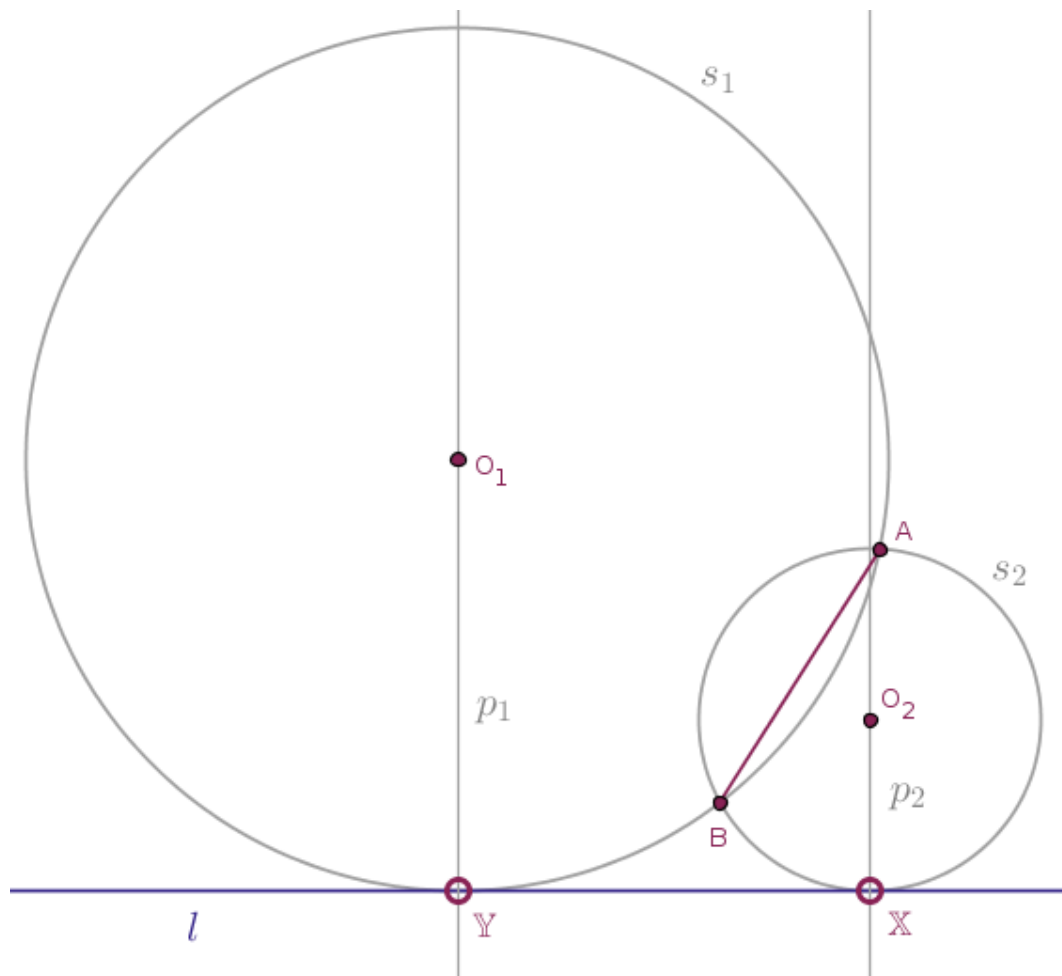
Alternatively, once the circles $s_{1,2}$ are obtained, draw a perpendicular bisector, b_{ab} , of the line segment AB until that bisector, b_{ab} , intersects the circle s_1 at the outer point C and the circle s_2 - at the outer point D (Figure 16.5):



Then, bisect the line segments CB and BD with the straight lines b_{cb} and b_{bd} correspondingly.

The center O_1 of the circle s_1 will be the point where the straight lines b_{ab} and b_{cb} intersect; the center O_2 of the circle s_2 will be the point where the straight lines b_{ab} and b_{bd} intersect.

Once the points $O_{1,2}$ are located, a straight line p_1 through the point O_1 perpendicular to the straight line l will intersect l at the candidate point $\mathbb{Y}$; a straight line p_2 through the point O_2 perpendicular to the straight line l will intersect l at the candidate point $\mathbb{X}$ (Figure 16.6):



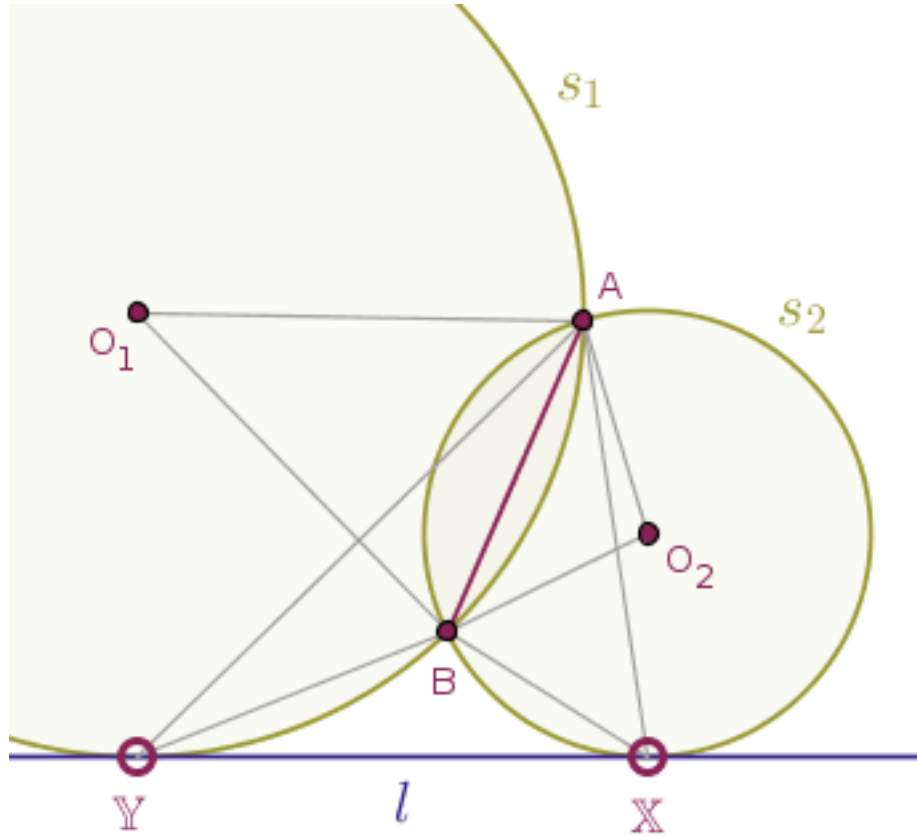
Coming back to the Generalized Angle Optimization Problem Of Regiomontanus, which one of the above two candidate points, X or Y , marks the spot?

The point of tangency of the given straight line, l , and **the smaller** circle or the circle with the smaller radius.

In other words, if r_1 is the radius of the solution circle s_1 and r_2 is the radius of the solution circle s_2 then:

- if $r_1 < r_2$ then the point Y marks the spot
- if $r_1 > r_2$ then the point X marks the spot
- if $r_1 = r_2$ then both points, X and Y , mark a spot

Evidently, in the first two cases above the solution is unique, while in the last case above it is not. In order to see that, recall that the angles under which the input line segment AB is seen from the candidate points X and Y correspondingly (Figure 16.7):



are the halves of their corresponding generating central angles:

$$\angle AXB = \frac{1}{2} \angle AO_2B, \quad \angle AYB = \frac{1}{2} \angle AO_1B$$

But, clearly, the point O_2 that is closer to the line segment AB , generates a larger angle than the point O_1 that is further away from the line segment AB :

$$\angle AO_2B > \angle AO_1B$$

Then, as we noted above, if $r_1 > r_2$ then the maximal angle problem has a unique solution, the point X generated by the smaller circle, $s_2(O_2, r_2)$, and the other way around and if $r_1 = r_2 = r$ then the maximal angle problem has two distinct solutions, each of which is generated by the circles $s_1(O_1, r)$ and $s_2(O_2, r)$ correspondingly, under the chosen configuration of the input objects.

We, of course, are on the hook to analyze the different outcomes of reflecting the given point B into its image point B' . Such an analysis, however, is not the purpose of this exercise. The purpose of this exercise is a plausible investigation of the psychology of an open-ended process of looking for an alternative solution and finding one.

But for the record, if the image point B' winds up strictly **inside** the circle l' then the problem has no solutions, if the image point B' winds up sitting **on** the circle l' then the problem has one solution, if the image point B' winds up strictly **outside** the circle l' then the problem has two distinct solutions. $\square$